SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Analytical Problem Solving (5 QUESTIONS)

Course Code:	MLA0407	**Course Title:**	**Deep Learning**
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – **Analytical problem solving**
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.***		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I ________________________ (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problem Solving **Duration: 1hr**

Q1.

Design a machine learning solution for predicting customer loan approval using a suitable learning algorithm. **Describe** the major stages involved in model development, including data preparation, model training, and performance evaluation. (10 marks)

Q2.
Describe the architecture of an Artificial Neural Network (ANN) consisting of input, hidden, and output layers. **Illustrate** the role of artificial neurons, weights, bias, and activation functions in processing input data to generate predictions. (10 marks)

Q3.
Analyze the working of the Perceptron learning algorithm for solving a binary classification problem. **Demonstrate** the weight and bias update process for one iteration using a suitable numerical example. (10 marks)

Q4.
Illustrate the forward propagation process in a Multilayer Perceptron (MLP) using a simple neural network with one hidden layer. **Compute** the output values for the given inputs, weights, biases, and activation function. (10 marks)

Q5.
Demonstrate the Backpropagation algorithm for training a neural network by calculating the output error, propagating the error through hidden layers, and updating the network weights using Gradient Descent. (10 marks)

Q6.
Compare and evaluate Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent with respect to convergence behavior, computational efficiency, memory requirements, and practical applications in deep learning. (10 marks)

Q7.
Formulate a Maximum Likelihood Estimation (MLE) approach for estimating the parameters of a probability distribution from a given dataset. **Interpret** the estimated parameters and discuss their significance in machine learning model development. (10 marks)

Q8.
Analyze the impact of the Curse of Dimensionality on machine learning models with respect to training complexity, prediction accuracy, and data sparsity. **Propose** suitable techniques to overcome these challenges in real-world applications. (10 marks)

Q9.
Design a Multilayer Perceptron (MLP) model for a multiclass image classification problem. **Describe** the network architecture and explain the role of forward propagation, backpropagation, and Gradient Descent during the training process. (10 marks)

Q10.
Evaluate the importance of learning algorithms in building intelligent machine learning

systems. **Compare** traditional machine learning approaches with neural network-based learning by considering model complexity, learning capability, and application scenarios. (10 marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________